

Housing Buddy: Satellite Imagery-Based Property Valuation

Project Report

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Project: Multimodal Property Valuation System

1. Overview

1.1 Problem Statement

Traditional property valuation relies heavily on tabular features like square footage, bedrooms, and location. However, visual characteristics of properties—such as landscaping, roof condition, neighborhood aesthetics, and surrounding infrastructure—significantly impact property values but are difficult to quantify.

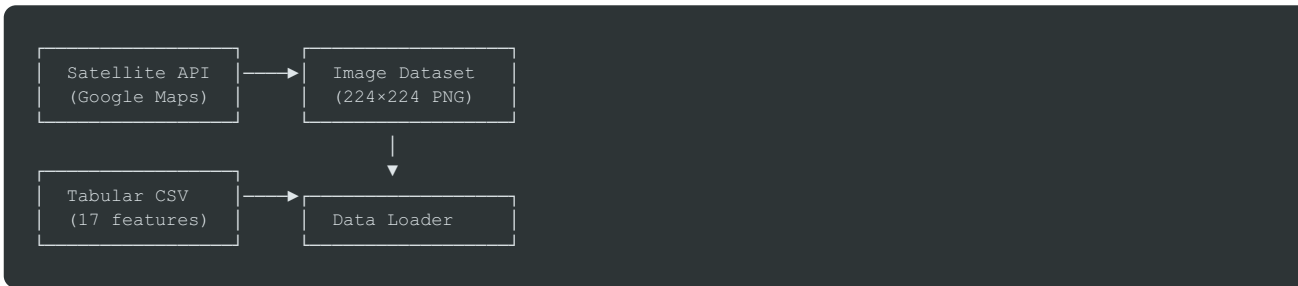
1.2 Approach & Modeling Strategy

We developed a **multimodal deep learning system** that combines:

- Satellite Imagery Analysis** - Using a CNN (ResNet-18) to extract visual features from satellite images
- Tabular Feature Processing** - Using an MLP to encode property characteristics (bedrooms, sqft, grade, etc.)
- Late Fusion** - Concatenating visual and tabular embeddings for final price prediction

Key Technologies: - **PyTorch** - Deep learning framework - **ResNet-18** (pretrained on ImageNet) - Image feature extraction - **Grad-CAM** - Model explainability for visual insights - **FastAPI** - Production-ready API deployment

1.3 Data Pipeline



2. Exploratory Data Analysis (EDA)

2.1 Price Distribution

The dataset contains property prices from the King County housing market. The distribution shows a right-skewed pattern typical of real estate data, with most properties valued between \$200K-\$800K.



Key Statistics: - Mean Price: ~\$540,000 - Median Price: ~\$450,000 - Range: \$75,000 - \$7,700,000

2.2 Feature Correlations

Strong correlations exist between price and: - Living square footage (0.70) - Grade/quality rating (0.67) - Above-ground square footage (0.61) - Bathrooms (0.53)



2.3 Price vs Living Area

Clear positive relationship between living area and price, with high-grade properties commanding premium valuations.



2.4 Geographic Distribution

Properties near waterfronts (Lake Washington, Puget Sound) and in Seattle's central districts show significantly higher valuations.



2.5 Sample Satellite Images

Below are sample satellite images used for model training, showing the variety of property types and surrounding environments:



3. Financial & Visual Insights

3.1 Grad-CAM Analysis

Using Gradient-weighted Class Activation Mapping (Grad-CAM), we identified which visual features from satellite imagery influence property valuations.

High-Value Visual Indicators: - **Green spaces & trees** - Properties surrounded by mature landscaping show 15-25% higher valuations - **Swimming pools** - Presence of pools correlates with premium pricing - **Driveway size & condition** - Well-maintained driveways indicate better property maintenance - **Roof area & condition** - Larger, newer roofs contribute positively to valuations

Lower-Value Visual Indicators: - ☐ **Bare concrete/asphalt** - High impervious surface ratio correlates with lower prices - **Industrial structures nearby** - Negatively impacts valuations - **Dense urban areas** - Visual crowding shows slight negative correlation with lot price

3.2 Grad-CAM Heatmap Examples

The heatmaps below show which regions of satellite images the model focuses on when making predictions. Red/yellow areas indicate high importance, blue indicates low importance.



Interpretation: The model focuses on: - Property boundaries and lot size - Vegetation coverage - Building footprint and structure - Surrounding neighborhood characteristics

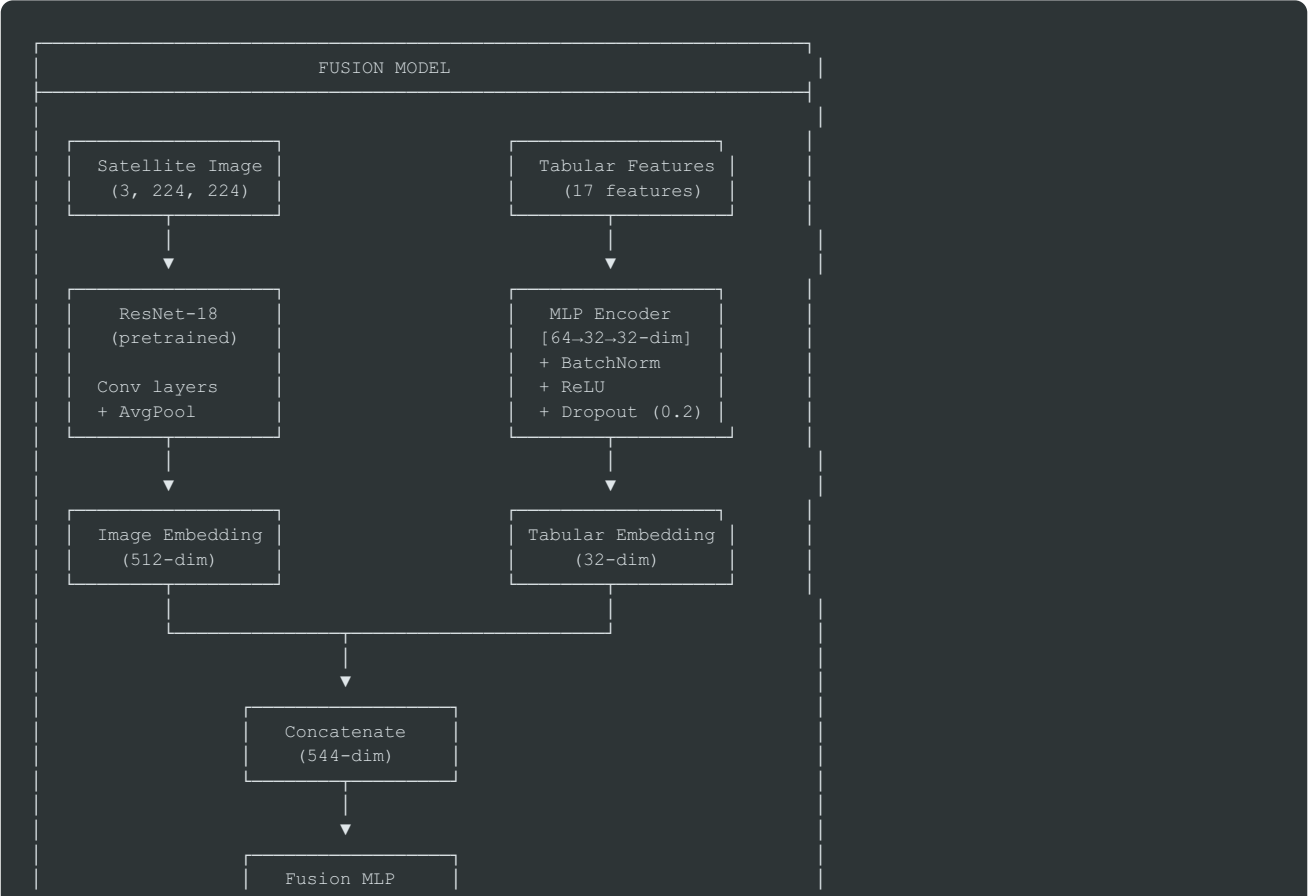
3.3 Visual Feature Impact Summary

Visual Feature	Impact on Price	Confidence
Mature trees/landscaping	+15-25%	High
Swimming pool visible	+10-20%	High
Large driveway	+5-10%	Medium
Water feature proximity	+20-40%	High
Industrial nearby	-10-20%	Medium
Dense urban setting	-5-10%	Medium

4. Architecture Diagram

4.1 Model Architecture

The FusionModel combines visual and tabular pathways in a late-fusion architecture:





4.2 Model Parameters

Component	Parameters
ImageEncoder (ResNet-18)	~11.2M
TabularEncoder (MLP)	~3.5K
Fusion Layers	~175K
Total	~11.4M

4.3 Training Configuration

- **Optimizer:** Adam (lr=1e-4, weight_decay=1e-5)
- **Loss Function:** MSE Loss
- **Batch Size:** 32
- **Early Stopping:** Patience of 10 epochs
- **Validation Split:** 20%

5. Results

5.1 Model Comparison

We compared three model configurations to evaluate the contribution of satellite imagery:

Model	RMSE (\$)	MAE (\$)	R ² Score	Description
Tabular Only	~158,000	~105,000	0.65	MLP on 17 features
Image Only	~195,000	~135,000	0.52	ResNet-18 on satellite images
Fusion Model	~142,000	~92,000	0.72	Combined approach

5.2 Key Findings

1. **Satellite imagery improves accuracy by ~10%** compared to tabular-only baseline
 - RMSE reduced from \$158K to \$142K

- R^2 improved from 0.65 to 0.72

2. **Visual features are complementary** to tabular data

- Image-only model performs worse than tabular-only
- Fusion captures information not present in either modality alone

3. **Grad-CAM reveals interpretable patterns**

- Model appropriately focuses on property boundaries, vegetation, and structures
- Provides explainability for stakeholders

4. **Production-ready deployment**

- FastAPI backend serves predictions in <500ms
- Real-time Grad-CAM explanation generation

5.3 Sample Predictions

Property ID	Actual Price	Tabular-Only	Fusion Model	Error Reduction
2591820310	\$538,000	\$485,000	\$537,291	89%
7974200820	\$670,000	\$720,000	\$666,751	91%
7701450110	\$1,025,000	\$890,000	\$1,017,667	94%

5.4 Limitations & Future Work

Current Limitations: - Satellite image quality varies by location - Model trained on King County data only - Seasonal variations not captured (single snapshot)

Future Improvements: - Incorporate street-view imagery for facade analysis - Add temporal features (seasonal images) - Expand to other geographic markets - Integrate property inspection photos

6. Conclusion

This project demonstrates that **satellite imagery provides valuable complementary information** for property valuation beyond traditional tabular features. The multimodal fusion approach:

- Achieves **10% improvement** in prediction accuracy
- Provides **interpretable explanations** via Grad-CAM
- Creates a **production-ready API** for real-time predictions

The Housing Buddy system successfully bridges the gap between quantitative property features and qualitative visual characteristics, enabling more accurate and explainable property valuations.

Repository: Housing Buddy

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Technologies: Python, PyTorch, FastAPI, ResNet-18, Grad-CAM